

Higher-Origin Holdout for Source-Native Masked L^2 Polarization: Finite Transfer and Floor-Stability Firewall

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Abstract

The source-native residual $\beta(t) = \Lambda(t+2) - b^{(2)}(t)$ was attached in TPC-353 to the literal divisibility-masked prime-shell operator on a low-origin panel. We test transfer to a disjoint higher-origin panel while keeping every other protocol choice fixed. For every finite row the identity $\|A\beta\|_2^2 = \|A\Lambda\|_2^2 + \|Ab\|_2^2 - 2\langle A\Lambda, Ab \rangle$ is exact, with a corresponding Cauchy envelope. On 216 higher-origin law-level rows, all four predeclared sign laws have positive output alignment. For the all-plus law the normalized alignment coefficient ranges from 0.6508 to 0.9914, compared with 0.6929 to 0.9963 on the locked TPC-353 parent; its mean drops by 0.02125. Positive transfer survives, but the finite floor is not stable. This is a scoped holdout result, not an asymptotic arithmetic estimate, power credit, or twin-prime conclusion.

1 Question and scope

TPC-352 subjected a reciprocal-shell finite repair to a disjoint holdout and found a high-shell floor below its balanced parent. TPC-353 then attached the actual source-native vector to the literal masked operator on a low-origin panel. The remaining local question is whether that operator-level alignment survives a higher-origin holdout when no response is used to retune the model. The answer is positive at the census level but loses a measurable all-plus floor and mean.

Throughout, all conclusions are explicitly classified as finite, declared model statements or numerically certified finite observations. In particular, the finite V59 convention used below is not identified with an asymptotic twin-prime formula.

2 Finite object and exact identities

Let I be a finite interval and let $S_Q = \{p : p \text{ prime}, Q < p \leq 2Q\}$. For a sign law e define the literal operator

$$A_e(u, t) = \sum_{p \in S_Q} e_p 1_{u \neq t} 1_{p \nmid ut} \frac{H^{2s}}{(H^2 + (u-t)^2)^s} \left(1_{u \equiv t \pmod{p}} - \frac{1}{p-1} \right). \quad (1)$$

Both endpoint masks in $1_{p \nmid ut}$ are retained. The source is the finite declared model

$$\beta = \Lambda - b, \quad b(t) = 2C_2, 1_{2 \nmid t} \prod_{p \mid t, p > 2} \frac{p-1}{p-2}, \quad (2)$$

with the inherited finite Euler-tail enclosure and logarithm midpoint rule.

Proposition 1 (operator polarization). For any real finite matrix A and finite vectors L, b , putting $\beta = L - b$ gives

$$\|A\beta\|_2^2 = \|AL\|_2^2 + \|Ab\|_2^2 - 2\langle AL, Ab \rangle. \quad (3)$$

This follows by expanding the finite Euclidean square; no limit or interchange is used.

Write $E_L = \|AL\|_2^2$, $E_b = \|Ab\|_2^2$, and, when their sum is nonzero,

$$\kappa_A = \frac{2\langle AL, Ab \rangle}{E_L + E_b}, \quad R_A = \frac{\|A\beta\|_2^2}{E_L + E_b}. \quad (4)$$

Then $R_A = 1 - \kappa_A$. Cauchy–Schwarz gives the exact finite envelope

$$\frac{(\sqrt{E_L} - \sqrt{E_b})^2}{E_L + E_b} \leq R_A \leq \frac{(\sqrt{E_L} + \sqrt{E_b})^2}{E_L + E_b}. \quad (5)$$

The envelope is an interface for a specified finite source and operator, not a source-uniform operator theorem.

3 Protocol and independent audit

The panel was fixed before evaluating responses. Relative to TPC-353, only the origins change:

parent origins	6001, 8001, 10001
holdout origins	21001, 23001, 25001
source counts	256, 512, 1024
shell anchors	$Q = 24, 54, 80$
kernel exponents	$s = 1, 2$
sign laws	all-plus, alternating-index, mod-4 character, half split
height and source cutoff	$H = 66, 50000$

The Cartesian product has $3 \cdot 3 \cdot 3 \cdot 2 \cdot 4 = 216$ holdout rows. The producer uses Decimal/Fraction source enclosures followed by a float64 matrix replay. A separate checker rebuilds the source with a trial sieve and accumulates shell primes in reverse order. It does not import the producer. Eight in-memory mutation tests verify the certificate firewall, and the certificate records min/mean/max transfer deltas against the hash-locked TPC-353 parent.

An exact fourteen-point anchor on [21001, 21014] and shell {5, 7} uses two fixed rational vectors. Its four reduced-fraction digests are independently recomputed, and equation (3) holds exactly. The largest floating point identity residual over the numerical panel is $1.6093254089355469 \times 10^{-6}$.

4 Results

Table 1 reports the output coefficient κ_A and the remaining residual fraction R_A . Every row is separated from zero by the declared 10^{-7} guard; no row is unresolved.

Table 1: Operator-level polarization on the higher-origin holdout (54 rows per law).

law	rows	$\kappa_{\min}$	$\kappa_{\max}$	$\bar{\kappa}$	R range
all-plus	54	.650760	.991350	.874362	.008650–.349240
alternating-index	54	.007731	.669776	.183032	.330224–.992269
mod-4 character	54	.021947	.713036	.320578	.286964–.978053
half split	54	.039348	.656130	.314469	.343870–.960652

The source coefficient, computed before applying A_e , lies in [0.3635760668, 0.3864841937] on the holdout rows. For all-plus, the output coefficient exceeds the source coefficient on every row, with output-minus- source range [0.2740398279, 0.6242474791]. For the other laws the deltas still range across both signs: alternating-index [−0.3656732564, 0.2847031469], mod-4 [−0.3584238082, 0.3279629836], and half split [−0.3339835447, 0.2846881223].

The locked-parent comparison gives the all-plus minimum shift −0.0421511462 and mean shift −0.0212497456. The holdout therefore preserves finite positive alignment while refuting a uniform all-plus floor transfer at this scope.

The output cosine for all-plus is 0.700035–0.991441; for the other laws the minima are respectively 0.029389, 0.097886, and 0.110696. These values make the geometry visible without replacing the exact identity: the masked operator can align the two source components much more strongly than their source-space inner product suggests.

5 Claim firewall and route decision

The exact claims are the finite operator identity and Cauchy envelope. The declared-model claim is the literal source attachment. The numerical claims are the 216-row higher-origin replay, positive alignment on 216/216, the origins-only comparison to TPC-353, and the ranges in Table 1. The narrow obstruction is that positive finite transfer does not stabilize the all-plus floor: the holdout minimum and mean are lower than the parent. Source-level polarization also does not determine operator-level polarization; consequently a source-only L^2 ledger cannot be promoted silently to a masked estimate.

No source-uniform arithmetic L^2 bound, uniform masked-operator bound, canonical sign law, strict 1/400 payment, fixed-power credit, Route-B reassembly, or twin-prime endpoint follows. The Session-named Route-A and Route-B evaluator files are absent from this checkout, so the local Bridge-B is fail-closed fallback evidence rather than an official evaluator pass. The next admissible test is a position-aware masked normalization or a controlled sign-law subspace; adding the same finite panel again would not pay the open theorem.

References

- [1] Standard finite-dimensional Cauchy–Schwarz and polarization identities are used only in the elementary form displayed in equations (3)–(5).